

FMP 2 - INSURANCE COMPANIES AND PENSION PLANS

- $P[\text{Survival until age } N + 1]$
 $= P[\text{Survival until age } N] \times [1 - \text{probability of death at age } N + 1]$

- The probability that a policyholder aged X will survive until age Y

$$= \frac{\text{Probability of Survival from birth until age } Y}{\text{Probability of Survival from birth until age } X}$$

Key ratios for P&C insurance companies –

- Loss ratio = ratio of payouts to premiums.
- Expense ratio = total expenses divided by premiums received
- Combined ratio = loss ratio + expense ratio
- Combined ratio after dividends =
combined ratio + dividends paid to the policyholders as a percentage of premiums
- Operating ratio =
combined ratio after dividends – investment income as a percentage of premium

FMP 3 - FUND MANAGEMENT

- Net asset value (NAV) represents a fund's per share market value.

$$NAV = \frac{\text{Fund Assets} - \text{Fund Liabilities}}{\text{Number of shares outstanding}}$$

FMP 4 –

INTRODUCTION TO DERIVATIVES

	Long Position Payoff	Short Position Payoff
Forward Contract	$S_T - K$	$K - S_T$
European Call Option	$\max(S_T - K, 0)$	$-\max(S_T - K, 0)$
European Put Option	$\max(K - S_T, 0)$	$-\max(K - S_T, 0)$

FMP 8 –

USING FUTURES FOR HEDGING

- Basis = spot price of asset – futures price of contract.*

Net price received when a short hedge is used = $F_0 + b_t$

Net cost of asset when long hedge is used = $F_0 + b_t$

where,

F_0 - Futures price at time t_0

F_t - Futures price at time t

b_t - Basis at time t ($S_t - F_t$)

- Optimal hedge ratio is given by -

$$h^* = \rho \times \frac{\sigma_S}{\sigma_F}$$

where,

ρ = coefficient of correlation between ΔS and ΔF

σ_S = Standard deviation of ΔS

σ_F = Standard deviation of ΔF

h^* = Optimal hedge ratio

- Optimal number of Futures contracts for hedging is defined as -

$$N^* = h^* \times \frac{Q_A}{Q_F}$$

where,

Q_A = Size of the position being hedged

Q_F = Number of units of the assets underlying one futures contract

N^* = Optimal number of contracts for hedge

- For a series of one-day hedges, the dollar value of the position being hedged (V_A) and the dollar value of one futures contract (V_F), is used. So, the number of contracts is given by -

$$N^* = h^* \times \frac{V_A}{V_F} = h^* \times \frac{Q_A}{Q_F} \times \frac{S}{F}$$

For a well-diversified equity portfolio stock index futures can be used to hedge the exposure, the optimal hedge ratio can be assumed to be **1.0**. Hence, the number of contracts to be shorted is given by -

$$N^* = \frac{V_A}{V_F} = \frac{\text{Portfolio Value}}{\text{Futures Price} \times \text{Contract Multiplier}}$$

where,

V_A = Current value of the portfolio

V_F = Current value of one futures contract (the futures price times multiplier).

- In order to change original beta (β) of the portfolio can be changed to a target beta (β^*), the number of contracts that should be used is given by -

$$N^* = (\beta^* - \beta) \frac{V_A}{V_F}$$

- When $\beta^* < \beta$, the sign of N^* is negative and we need to short the contracts.
- When $\beta^* > \beta$, the sign of N^* is positive and we need to go long in the contracts.

FMP 9 –

FOREIGN EXCHANGE MARKETS

- Real interest rate is defined as -

$$R_{real} = \frac{1 + R_{nom}}{1 + R_{infl}} - 1$$

where,

R_{real} is the real interest rate

R_{nom} is the nominal interest rate

R_{infl} is the rate of inflation.

This is often approximated as -

$$R_{real} = R_{nom} - R_{infl}$$

- In absence of any arbitrage opportunities, for an exchange rate of **XXXXYY**, the Covered Interest Parity gives the relationship for forward exchange rate as -

$$F = S \frac{(1 + R_{YY})^T}{(1 + R_{XX})^T}$$

where,

F = Forward exchange rate at time T

S = Current exchange rate

R_{XX} = Foreign interest rate

R_{YY} = Domestic interest rate

T = Time for maturity of forward contract.

- It is approximately true that when $T < 1$, then forward rate points can be given as -

$$\frac{F - S}{S} = (R_{YY} - R_{XX})T$$

where,

$F - S$ is expressed as points divided by 10,000.

FMP 10 –

PRICING FINANCIAL FORWARDS AND FUTURES

- The relationship between the forward price and the spot price of various types of assets -

Type of Financial Asset	Theoretical value of Forward Price
No Income Asset	$F_0 = S_0 \times (1 + R)^T$
Known Income Asset	$F_0 = (S_0 - I) \times (1 + R)^T$
Asset with Known Yield	$F_0 = S_0 \left(\frac{1 + R}{1 + Q} \right)^T$

- For all the above cases -
 - If *Actual value of F_0* > *Theoretical Value of F_0* , then arbitrageurs can buy the asset and short forward contracts on the asset.
 - If *Actual value of F_0* < *Theoretical Value of F_0* , they can short the asset and enter into long forward contracts on it.
- Value of a forward contract for different classes of financial assets -

Type of Asset	Value of Long Position	Value of Short Position
Asset with No Income	$S - \frac{K}{(1 + R)^T} = \frac{F - K}{(1 + R)^T}$	-(Value of Long Position)
Asset with Known Income	$(S - I) - \frac{K}{(1 + R)^T} = \frac{F - K}{(1 + R)^T}$	
Asset with Known Yield	$\frac{S_0}{(1 + Q)^T} - \frac{K}{(1 + R)^T} = \frac{F - K}{(1 + R)^T}$	

where,

F is the current forward price

S is the current spot price of the asset

R is the risk-free rate of interest

Q is the dividend yield

T is the time to maturity

I is the present value of discrete dividends (payable before the maturity of the contract)

K is the forward price for a contract initiated some time ago.

It is quite clear that in all three cases, value of long position is equal to

$$\frac{F - K}{(1 + R)^T}$$

FMP 11 – COMMODITY FORWARDS AND FUTURES

- For commodities held exclusively for investment purpose, the formulae for financial assets can be applied directly to calculate the forward prices.
- If the lease rate for an investment commodity is l , the relationship between the forward price F_0 and the spot price S_0 is given as

$$F_0 = S_0 \times \left(\frac{1+R}{1+l} \right)^T \quad \text{Or,} \quad l = \left(\frac{S_0}{F_0} \right)^{\frac{1}{T}} (1+R) - 1$$

- If the convenience yield associated with an investment commodity is Y , the forward price is given as

$$F_0 = (S + U) \left(\frac{1+R}{1+Y} \right)^T \quad \text{Or,} \quad Y = \left(\frac{S+U}{F_0} \right)^{\frac{1}{T}} (1+R) - 1$$

- For commodities that are held almost always for consumption purposes, the formulae for financial assets can be used to calculate the upper bound of the forward prices.

- The upper bound to the forward price is defined as -

$$F_0 \leq S_0(1+R)^T$$

- When storage costs with present value U are considered, the upper bound to the forward price is defined as -

$$F_0 \leq (S_0 + U)(1+R)^T$$

- The cost of carry per year is defined as -

$$\frac{1+R}{1+Q} - 1$$

where, R is the financing cost and Q is the yield (all rates compounded annually)

This can also be approximated as $R - Q$ (this is accurate when R and Q are continuously compounded)

- When R and Q are continuously compounded, the forward price is then expressed as -

$$F_0 = S_0 \times e^{(R-Q) \times T}$$

- When the storage cost is expressed with continuous compounding, the cost of carry is the interest rate plus the storage cost. The relationship between forward price and the spot price is expressed as -

$$F_0 = S_0 \times e^{(C-Y) \times T}$$

where, C is the cost of carry and Y is the convenience yield.

- The expected future spot price of an asset is defined as -

$$E(S_T) = F \left(\frac{(1+X)^T}{(1+R)^T} \right)$$

where, X is the expected return and R is the risk-free rate.

FMP 12- OPTIONS MARKETS

Position	Payoff	Profit
Long Call	$\max(S_T - K, 0)$	$\max(S_T - K, 0) - c_0$
Short Call	$-\max(S_T - K, 0)$	$c_0 - \max(S_T - K, 0)$
Long Put	$\max(K - S_T, 0)$	$\max(K - S_T, 0) - p_0$
Short Put	$-\max(K - S_T, 0)$	$p_0 - \max(K - S_T, 0)$

FMP 13 – PROPERTIES OF OPTIONS

- For prices of European call Options -

Type of asset	Lower Bound	Upper Bound
Non-Dividend Paying Stock	$\max(S_0 - PV(K), 0)$	S_0
Dividend Paying stock	$\max(S_0 - PV(D) - PV(K), 0)$	S_0
Assets other than Stocks	$PV(F) - PV(K)$	$PV(F)$

- For prices of European Put Options -

Type of Asset	Lower Bound	Upper Bound
Non-Dividend Paying Stock	$\max(PV(K) - S_0, 0)$	S_0
Dividend Paying stock	$\max(PV(D) + PV(K) - S_0, 0)$	S_0
Assets other than Stocks	$PV(K) - PV(F)$	$PV(F)$

- For American Call Options -

Type of Asset	Lower Bound	Upper Bound
Non-Dividend Paying stock	$\max(S_0 - PV(K), 0)$	S_0
Dividend Paying Stock	--	S_0

- For American Put Options -

Type of Asset	Lower Bound	Upper Bound
Non-Dividend Paying stock	$\max(K - S_0, 0)$	K
Dividend Paying Stock	--	K

- Put-Call Parity for European Options -

Type of Asset	European Options
Non-Dividend Paying Stock	$c + PV(K) = p + S_0$
Dividend Paying Stock	$c + PV(D) + PV(K) = p + S_0$
Assets other than Stocks	$c + PV(K) = p + PV(F)$

- Put-Call Inequality for American Options -

Type of Asset	American Options
Non-Dividend Paying Stock	$S_0 - K \leq C - P \leq S_0 - PV(K)$
Dividend Paying Stock	$S_0 - PV(D) - K \leq C - P \leq S_0 - PV(K)$

- An American call option can be exercised on ex-dividend dates, if and only if, the dividend is less than,
 $K - K^*$

Where, K^* is defined as the present value of K , with discounting being done at the risk-free rate from the next ex-dividend date to the current ex-dividend date. (or option maturity if there are no further ex-dividend dates).

FMP 14 – TRADING STRATEGIES

- For this chapter, focus on understanding the strategies, and not on remembering the formulae, which have been presented here just for convenience.*
- The various ways in which an option position can be combined with a position in the asset, can be represented by the following strategies and the corresponding equations

Protective Put Strategy	$P + S = C + PV(K)$
Covered Call Strategy	$S - C = PV(K) - P$
Reverse protective put strategy	$-P - S = -C - PV(K)$
Reverse Covered Call Strategy	$C - S = P - PV(K)$

- Payoff from a bull spread comprising of call options can be illustrated as follows -

Range for S_T	Payoff from long call with strike Price K_1	Payoff from Short Call with Strike Price K_2	Total Payoff
$S_T \leq K_1$	0	0	0
$K_1 < S_T \leq K_2$	$S_T - K_1$	0	$S_T - K_1$
$S_T \geq K_2$	$S_T - K_1$	$-(S_T - K_2)$	$K_2 - K_1$

- Payoff from a bull spread comprising of put options can be illustrated as follows -

Range for S_T	Payoff from Long Put with Strike Price K_1	Payoff from Short Put with Strike Price K_2	Cash Amount	Total Payoff
$S_T \leq K_1$	$K_1 - S_T$	$-(K_2 - S_T)$	$K_2 - K_1$	0
$K_1 < S_T \leq K_2$	0	$-(K_2 - S_T)$	$K_2 - K_1$	$S_T - K_1$
$S_T > K_2$	0	0	$K_2 - K_1$	$K_2 - K_1$

- Payoff from a bear spread comprising of put options can be illustrated as follows -

Range for S_T	Payoff from Long Put with Strike Price K_2	Payoff from Short Put with Strike Price K_1	Total Payoff
$S_T \leq K_1$	$K_2 - S_T$	$-(K_1 - S_T)$	$K_2 - K_1$
$K_1 < S_T \leq K_2$	$(K_2 - S_T)$	0	$K_2 - S_T$
$S_T > K_2$	0	0	0

- Payoff from a bear spread comprising of call options can be illustrated as follows -

Range for S_T	Payoff from Long Call with Strike Price K_2	Payoff from Short Call with Strike Price K_1	Cash Amount	Total Payoff
$S_T \leq K_1$	0	0	$K_2 - K_1$	$K_2 - K_1$
$K_1 < S_T \leq K_2$	0	$-(S_T - K_1)$	$K_2 - K_1$	$K_2 - S_T$
$S_T > K_2$	$S_T - K_2$	$-(S_T - K_1)$	$K_2 - K_1$	0

- Payoff from a box spread can be illustrated as follows -

Range for S_T	Payoff from Bull Spread	Payoff from Bear spread	Total Payoff
$S_T \leq K_1$	0	$K_2 - K_1$	$K_2 - K_1$
$K_1 < S_T \leq K_2$	$(S_T - K_1)$	$(K_2 - S_T)$	$K_2 - K_1$
$S_T > K_2$	$(K_2 - K_1)$	0	$K_2 - K_1$

- Payoff from a butterfly spread comprising of call options can be illustrated as follows -

Range for S_T	Payoff from Long Call with Strike Price K_1	Payoff from Two Short Calls with Strike Price K_2	Payoff from Long Call with Strike Price K_3	Total Payoff
$S_T \leq K_1$	0	0	0	0
$K_1 < S_T \leq K_2$	$S_T - K_1$	0	0	$S_T - K_1$
$K_2 < S_T \leq K_3$	$S_T - K_1$	$-2(S_T - K_2)$	0	$2K_2 - K_1 - S_T = K_3 - S_T$
$S_T > K_3$	$S_T - K_1$	$-2(S_T - K_2)$	$S_T - K_3$	$2K_2 - K_1 - K_3 = 0$

- Payoff from a butterfly spread comprising of put options can be illustrated as follows -

Range for S_T	Payoff from Long Put with Strike Price K_1	Payoff from Two Short Puts with Strike Price K_2	Payoff from Long Put with Strike Price K_3	Total Payoff
$S_T \leq K_1$	$K_1 - S_T$	$-2(K_2 - S_T)$	$K_3 - S_T$	$2K_2 - K_1 - K_3 = 0$
$K_1 < S_T \leq K_2$	0	$-2(K_2 - S_T)$	$K_3 - S_T$	$S_T + K_3 - 2K_2 = S_T - K_1$
$K_2 < S_T \leq K_3$	0	0	$K_3 - S_T$	$K_3 - S_T$
$S_T > K_3$	0	0	0	0

- Payoff from a straddle can be illustrated as follows -

Range for S_T	Payoff from Long Call with Strike Price K	Payoff from Long Put with Strike Price K	Total Payoff
$S_T \leq K$	0	$K - S_T$	$K - S_T$
$S_T > K$	$S_T - K$	0	$S_T - K$

- Payoff from a strangle can be illustrated as follows -

Range for S_T	Payoff from Long Call with Strike Price K_2	Payoff from Long Put with Strike Price K_1	Total Payoff
$S_T \leq K_1$	0	$K_1 - S_T$	$K_1 - S_T$
$K_1 < S_T \leq K_2$	0	0	0
$S_T > K_2$	$S_T - K_2$	0	$S_T - K_2$

FMP 15 –

EXOTIC OPTIONS

- When a European call option is converted to a zero-cost product, it has a payoff of -

$$\max(S_T - K - A, -A)$$

Where $A = c(1 + R)^T$.

- Payoff from a Gap Call and Gap Put options can be illustrated as follows -

Range for S_T	Payoff from Gap Call Option	Payoff from Gap Put Option
$S_T \leq K_1$	0	$K_1 - S_T$
$K_1 < S_T \leq K_2$	0	0
$S_T > K_2$	$S_T - K_2$	0

- Payoffs from four types of Asian options are (if K is the strike price, S_T is the final asset price and S_{ave} is the average asset price) -

Type of Option	Payoff from Call	Payoff from Put
Average Price Option	$\max(S_{ave} - K, 0)$	$\max(K - S_{ave}, 0)$
Average Strike Option	$\max(S_T - S_{ave}, 0)$	$\max(S_{ave} - S_T, 0)$

- Payoffs from lookback options are (if K is the strike price, S_T is the final asset price, S_{min} and S_{max} are the minimum and maximum asset price in a given time respectively) -

Type of Option	Payoff from Call	Payoff from Put
Floating Lookback	$\max(S_T - S_{min}, 0)$	$\max(S_{max} - S_T, 0)$
Fixed Lookback	$\max(S_{max} - K, 0)$	$\max(K - S_{min}, 0)$

- Payoff from a volatility Swap and Variance Swaps are given as -

Type of SWAP	When Trader Pays Fixed	When Trader Pays Floating
Volatility Swaps	$L_{vol}(\sigma - \sigma_K)$	$L_{vol}(\sigma_K - \sigma)$
Variance Swaps	$L_{var}(\sigma^2 - V_K)$	$L_{var}(V_K - \sigma^2)$

- For Volatility Swap, L_{vol} is the principal, σ_K is the pre-specified yearly volatility and σ is the realized yearly volatility.
- For Variance Swap, L_{var} is the principal and V_K is the specified fixed variance and σ^2 is the realized variance.
- Sometimes, L_{vol} and σ_K are specified in a variance swap with the understanding that $V_K = \sigma_K^2$ and

$$L_{var} = \frac{L_{vol}}{2 \times \sigma_K}$$

FMP 16 - PROPERTIES OF INTEREST RATES

- The general formula for the future value (FV) of an amount (PV) invested at rate R for T years when R is expressed with compounding frequency of m times per year is -

$$FV = PV \left(1 + \frac{R}{m}\right)^{mT} \quad \text{Or,} \quad PV = FV \left(1 + \frac{R}{m}\right)^{-m}$$

With continuous compounding, the future value can be expressed as -

$$FV = PV \times e^{RT} \quad \text{Or,} \quad PV = FV \times e^{-RT}$$

- If R_1 is an interest rate when compounding frequency is m_1 and R_2 is the equivalent rate when compounding frequency is m_2 , then the relationship between R_1 and R_2 can be expressed as -

$$R_2 = m_2 \left[\left(1 + \frac{R_1}{m_1}\right)^{\frac{m_1}{m_2}} - 1 \right]$$

- If R_m is an interest rate expressed with compounding m and R_c is the same rate expressed with continuous compounding, then the relationship between R_m and R_c can be expressed as -

$$R_m = m \left(e^{\frac{R_c}{m}} - 1 \right) \quad \text{Or,} \quad R_c = m \times \ln \left(1 + \frac{R_m}{m} \right)$$

- If Y is the bond yield (with semi-annual compounding) for a bond paying coupons (c) 2 times every year and maturing in T years, then -

$$PV(\text{Bond}) = \frac{c}{2} \sum_{i=1}^{2T} \frac{1}{\left(1 + \frac{Y}{2}\right)^i} + \frac{100}{\left(1 + \frac{Y}{2}\right)^{2T}}$$

- The par yield of a bond, maturing in T years and paying coupons with a frequency of m times a year, can be expressed as -

$$Par = \frac{(100 - 100 \times d_T) \times m}{A(T)}$$

where,

d_T is the value of \$1 received at bond maturity (or the last discount factor)

$A(T)$ is the value of an annuity that pays \$1 on each coupon payment date. (or the sum of all discount factors)

- The approximate duration relationship is given as -

$$\Delta B = -DB\Delta Y$$

where,

B is the price of the bond.

D is the bond's duration.

ΔY is the change in the bond's yield.

ΔB is the change in the price resulting from the change in the yield.

- If the rates are compounded continuously, the Macaulay duration of the bond is defined as -

$$D = \sum_{i=1}^n \frac{PV_i}{PV_B} \times T_i$$

where, PV_i is the present value of the i^{th} cash flow. PV_B is the present value of the bond. T_i is the time in years for i^{th} cash flow.

- Modified Duration of the bond is defined as -

$$\text{Modified Duration} = \frac{\text{Macaulay Duration}}{1 + \frac{y}{m}}$$

where, y is the yield of the bond compounding m times per year.

- If dollar duration is $D_{\$}$, then $B = -D_{\$}\Delta y$

- If C is the convexity, the approximate relationship is given as -

$$\Delta B = -DB\Delta Y + \frac{1}{2}CB(\Delta Y)^2$$

where, C is the convexity of the bond.

- The Convexity of the bond is defined as -

$$C = \sum_{i=1}^n \frac{PV_i}{PV_B} \times T_i^2$$

where,

PV_i is the present value of the i^{th} cash flow,

PV_B is the present value of the bond,

T_i is the time in years for i^{th} cash flow.

- If rates are discretely compounded, then forward rate is given by

$$F = \left[\frac{(1 + R_2)^{T_2}}{(1 + R_2)^{T_1}} \right]^{\frac{1}{T_2 - T_1}} - 1$$

- If rates are continuously compounded, then forward rate is given by -

$$F = \frac{R_2 \times T_2 - R_1 \times T_1}{T_2 - T_1}$$

- The payoff and value of an **FRA** is as shown in the table -

Leg	Payoff from FRA	Value of FRA
Pay fixed and received floating	$\frac{(R - R_K)\tau L}{1 + R_\tau}$	$PV \left(\frac{(R_F - R_K)\tau L}{1 + R_F \tau} \right)$
Pay floating and received fixed	$\frac{(R_K - R)\tau L}{1 + R_\tau}$	$PV \left(\frac{(R_K - R_F)\tau L}{1 + R_F \tau} \right)$

where,

R is the realized floating rate

R_K is the fixed rate

L is the principal to which the rates are applied

τ is the length of the time period

R_F is the forward rate for the period underlying the **FRA**

PV is the present value the **FRA** today.

FMP 17- CORPORATE BONDS

- The expected loss rate on a bond in a given year can be defined as:

$$[\text{Probability of Default} \times (1 - \text{Expected Recovery Rate})]$$

- The expected return from a bond is,

$$[\text{Risk} - \text{Free Rate} + \text{Credit Spread} - \text{Expected Loss Rate}]$$

FMP 18 –

MORTGAGES AND MORTGAGE-BACKED SECURITIES

- The relationship between the amount borrowed A for time period T with the interest rate R (compounded monthly), and the monthly payment X is -

$$\frac{X}{R/12} \left[1 - \frac{1}{\left(1 + \frac{R}{12}\right)^{12T}} \right] = A$$

- The weighted-average coupon (WAC) and is the weighted-average maturity (WAM) defined as -

$$WAC = \sum_{i=1}^n w_i c_i \quad \text{And,} \quad WAM = \sum_{i=1}^n w_i L_i$$

where,

n is the number of mortgages in the pool.

c_i is the coupon of the i^{th} mortgage.

L_i is the remaining life of the i^{th} mortgage.

$w_i = \frac{P_i}{\sum_{i=1}^n P_i}$ where, P_i is the remaining principal of the i^{th} mortgage.

- The CPR is calculated as -

$$CPR = 1 - (1 - SMM)^{12}$$

where,

SMM is the constant prepayment rate (conditional prepayment rate).

- The value of the roll is calculated as -

$$A - B + C - D$$

where,

A is the price at which the pool is sold during the first month (including accrued interest).

B is the price at which the pool is purchased during the second month (including accrued interest).

C is the interest earned on the proceeds of the sale for one month.

D is the coupon and the principal repayment that would have been received on the pool sold during the first month.

- A simple incentive function I for a pool could be -

$$I = WAC - R$$

where, WAC is the weighted-average coupon. R is the current mortgage rate available to borrowers.

A slightly more elaborate incentive function is -

$$I = (WAC - R) \times ALS \times A - K$$

where,

ALS is the average loan size.

K is the estimated cost of refinancing a loan.

A is an annuity factor

- The option-adjusted spread (OAS) is defined as -

$$OAS = \text{Expected MBS Return} - \text{Return on Treasury Instruments}$$

FMP 19 – INTEREST RATE FUTURES

- Accrued Interest (AI) - The interest earned between two dates is

$$\frac{\text{Number of days between dates}}{\text{Number of days in reference period}} \times \text{Interest earned in the reference period}$$
- To calculate AI, three popular day count conventions for bonds are -
 - *actual/actual*. (Treasury Bonds)
 - *30/360* (Corporate and Municipal Bonds)
 - *actual/360*. (Treasury Bills/Money Market Instruments)
- The dirty price is calculated by adding accrued interest to the clean price -

$$\text{Dirty Price} = \text{Clean Price} + \text{Accrued interest since the last coupon date}$$
- If Q is the quote and C is the corresponding cash price per $USD\ 100$ of face value -

$$Q = \frac{360}{n} (100 - C) \quad \text{Or,} \quad C = 100 - \frac{nQ}{360}$$
- The price received for a delivered bond is determined (in part) by what is called the conversion factor. Thus, the price received for a bond is -

$$Sf + A$$

where,
 S is the most recent settlement price in the futures contract.
 f is the bond's conversion factor.
 A is the accrued interest calculated on an *actual/actual* basis.
- The cheapest-to-deliver bond is the one for which $Q - Sf$ is the least.

where,
 S is the most recent settlement price in the futures contract. f is the conversion factor, Q is the quoted price of the bond.
- The Treasury bond futures contract is simply a futures contract on an asset providing known income and the futures price is -

$$(S - I)(1 + R)^T$$

where,
 I is the present value of the coupons that will be received during the life of the futures contract,
 S is the current price of the bond,
 T is the life of the futures contract, and
 R is the (annually compounded) risk-free rate for maturity T .

If R is expressed with continuous compounding, the futures price becomes -

$$(S - I)e^{RT}$$
- Eurodollar Futures - The contract price is defined as $10,000 \times [100 - 0.25 \times (100 - Q)]$

where Q is the quote.

- Convexity Adjustment - $\text{Forward rate} = \text{Futures rate} - \frac{1}{2}\sigma^2 T_1 T_2$
- The number of long contracts that should be traded (with a negative number indicating a short position) for a hedge should be given as -

$$\frac{E_V}{E_F}$$

where,

E_F : The increase in value of one futures contract for a 1-basis point downward parallel shift in the zero curve.

E_V : The increase in value of a trader's position for a 1-basis point downward parallel shift in the zero curve.

- Duration-based hedge ratio / price sensitivity hedge ratio.

$$N^* = \frac{PD_P}{V_F D_F}$$

where

V_F is the contract price for one interest rate futures contract

D_F is the duration of the asset underlying the futures contract at the maturity of the futures contract

P is the forward value of the portfolio being hedged at the maturity of the hedge (in practice, this is usually assumed to be the same as the value of the portfolio today)

D_P is the duration of the portfolio at the maturity of the hedge.

FMP 20 – SWAPS

- Comparative advantage - total apparent gain from swap arrangement is $|\Delta_{fixed} - \Delta_{floating}|$, where Δ_{fixed} is the difference between the interest rates facing the two companies in fixed-rate markets, and $\Delta_{floating}$ is the difference between the interest rates facing the two companies in floating-rate markets.
- The position of the two counterparties can be considered as long in one type of bond and short in the other.
 - To the pay fixed, receive floating party, $V_{swap} = B_{floating} - B_{fixed}$
 - To the pay floating, receive fixed party, $V_{swap} = B_{fixed} - B_{floating}$
- Value of swap where dollars are received and a foreign currency is paid, is

$$V_{swap} = B_D - S_0 B_F$$

where B_F is the value, measured in the foreign currency, of the bond defined by the foreign cash flows on the swap and B_D is the value of the bond defined by the domestic cash flows on the swap, and S_0 is the spot exchange rate (expressed as number of dollars per unit of foreign currency).

- Similarly, the value of a swap where the foreign currency is received and dollars are paid is

$$V_{swap} = S_0 B_F - B_D$$